Ch 07: Lists: Sequences and Slicing

Application Program Development

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Sequences

- **Sequence**: an object that contains multiple items of data
  - The items are stored in sequence one after another
  - Items can be accessed by integer index, first item in index 0, second at 1, etc.
- Python provides different types of sequences, including **lists** and **tuples**
  - The difference between these is that a list is **mutable** (modifiable)
  - and a tuple is **immutable** (read-only)

Sequence

A sequence is an object that holds multiple items of data, stored one after the other. You can perform operations on a sequence to examine and manipulate the items stored in it.



Introduction to Lists

Lists

A list is an object that contains multiple data items. Lists are **mutable**, which means that their contents can be changed during a program's executions. Lists are dynamic data structures, meaning items may be added to them or removed from them. You can use indexing, slicing and various methods to work with lists in a program.

- **List**: an object that contains multiple data items (a **container**)
  - **Element**: an item in a list
  - Format: `list = [item1, item2, etc.]`
  - Can hold items of different types (compare to an array in C/Java)
- `print()` function can be used to display an entire list
- `list()` function can convert certain types of objects to lists

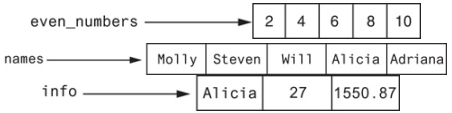


Figure 1: Lists of integers, strings and different types (Gaddis, 2021, pg.384)

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The Repetition Operator and Iterating over a List

- **Repetition operator**: makes multiple copies of a list and joins them together
  - The * symbol is repetition operator when applied to a sequence and an integer
    - Sequence is left operand, number is right
  - General format: `list * n`
- You can iterate over a list using a `for` loop
  - Format: `for x in list:`

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Indexing

- **Index**: a number specifying the position of an element in a list
  - Enables access to individual element in list
  - Index of first element in the list is `0`, second is `1`, and `n`'th element is `n-1`
  - Negative indexes identify positions relative to end of list
    - The index `-1` identifies the last element, `-2` the next to last, etc.

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The `len()` Function

- An `IndexError` exception is raised if an invalid index is used
- `len()` function: returns the length of a sequence such as a list
  - Example: `size = len(my_list)`
  - Returns the number of elements in the list, so the index of the last element is `len(list) - 1`
  - Can be used to prevent an `IndexError` exception when iterating over a list with a loop

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Lists are Mutable

- **Mutable sequence**: the items in the sequence can be changed
  - Lists are mutable, and so their elements can be changed
- An expression such as:

```
list[1] = new_value
```

can be used to assign a new value to a list

- Must use a valid index to prevent raising of an `IndexError` exception

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Concatenating Lists

- **Concatenate**: join two things together
- The + operator can be used to concatenate two lists
  - Cannot concatenate a list with another data type, such as a number
- The += augmented assignment operator can also be used to concatenate lists

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List Slicing

- **Slice**: a span of items that are taken from a sequence
  - List slicing format: `list[start : end]`
  - Span is a list containing copies of elements from `start` up to, but not including `end`
    - If `start` is not specified, 0 is used for start index
    - If `end` is not specified, `len(list)` is used for end index
  - Slicing expressions can include a step value and negative indexes relative to end of list



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Finding Items in Lists with the `in` Operator

- You can use the `in` operator to determine whether an item is contained in a list
  - General format: `item in list`
  - Returns `True` if the item is in the list, or `False` if it is not in the list
- Similarly you can use the `not in` operator to determine whether an item is not in a list

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Summary

- A list is a sequence, it contains multiple data items
- A list is a mutable sequence, elements can be changed and it can be appended to and have items removed from it
- The items of a list can be indexed by position
  - 0 based indexing, first item is at index 0, second at 1, and nth item at `n-1`
  - Can use negative index to indicate position relative from end of list
- List slicing is a powerful syntactic mechanism that allows you to get a copy of a span of items taken from a sequence
- The `in` and `not in` operator can be used to search for an item in a list



References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

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